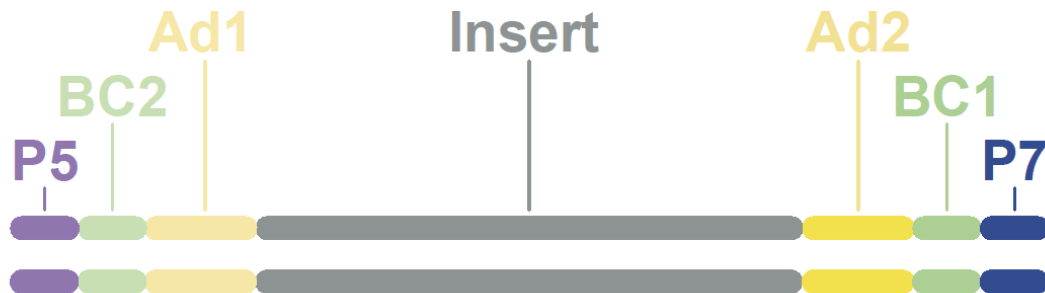


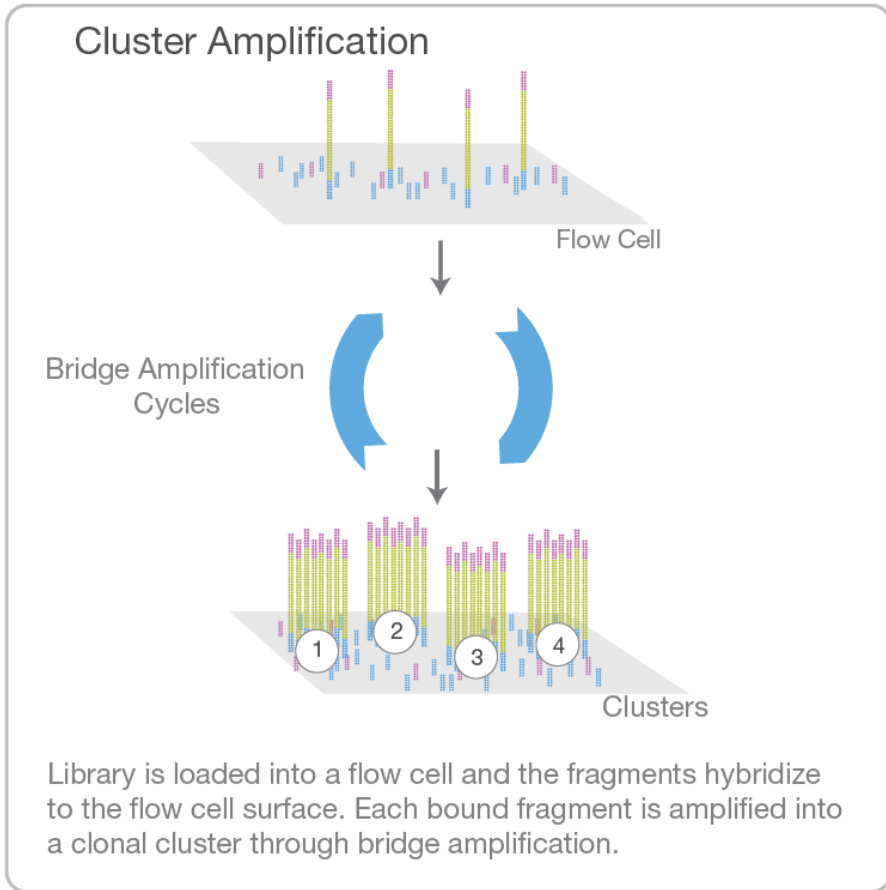
Library prep QC

Library prep - final library

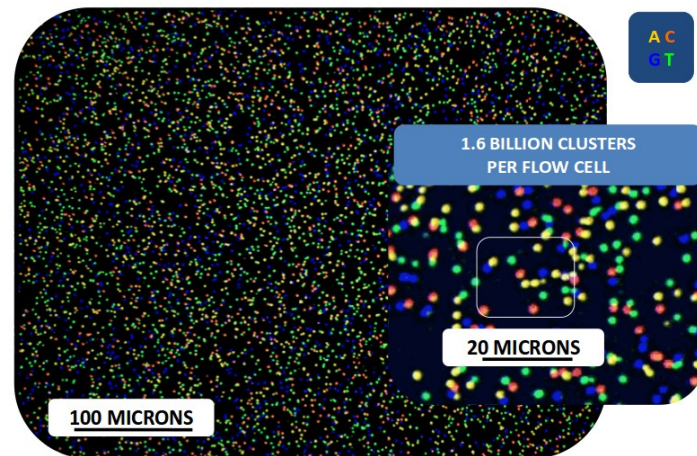


- Insert** = DNA fragment to be sequenced
- Ad1/Ad2** = Adapters that allow initiation of sequencing
- BC1/BC2** = Barcodes specific for each sample (one for each strand)
- P5/P7** = Flow cell binding sites

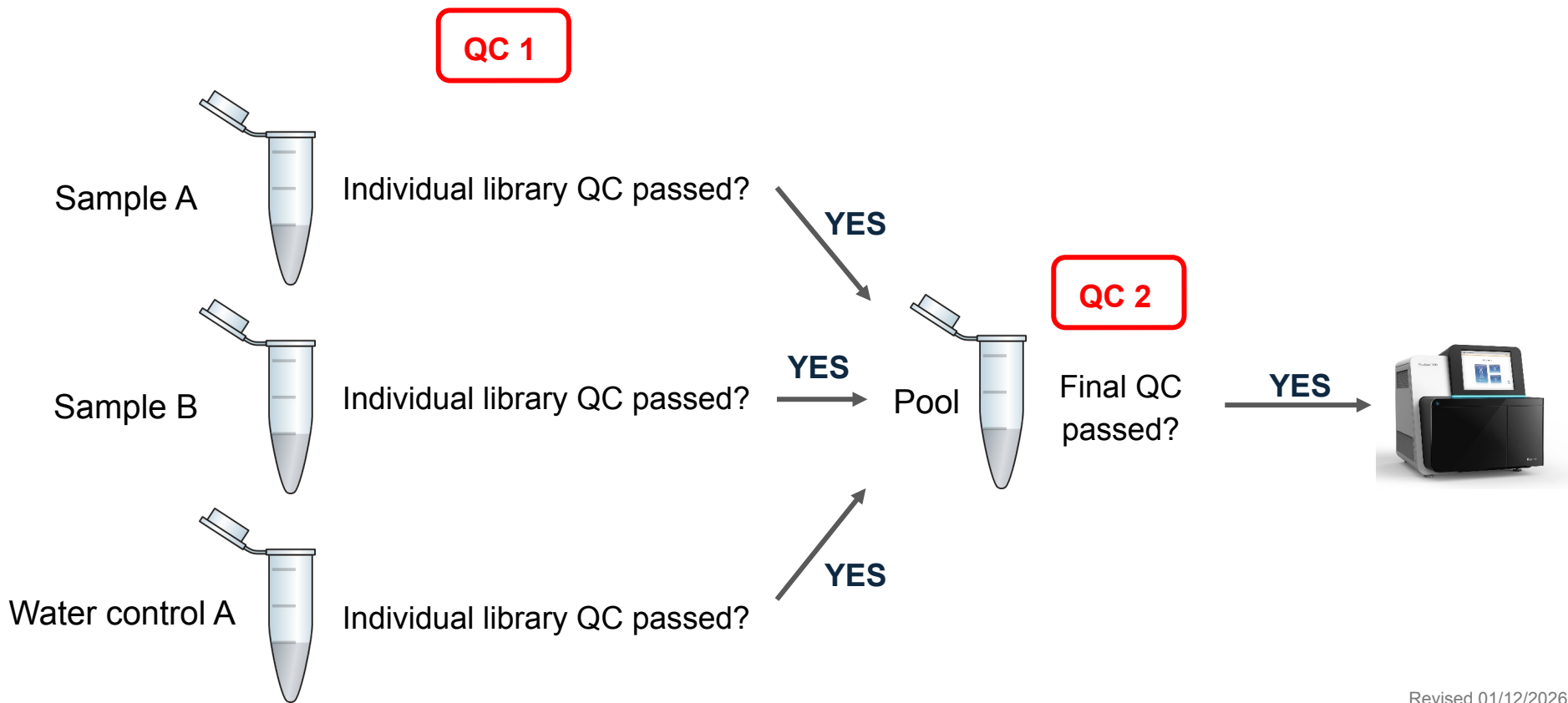
Cluster amplification is essential for Illumina sequencing



- ~ loading concentration
- ~ fragment length

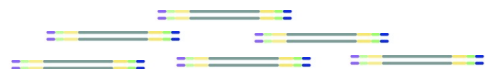


Library prep QC

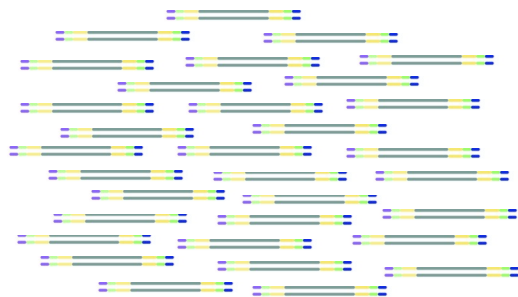


Library prep QC

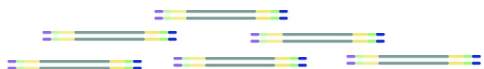
Concentration



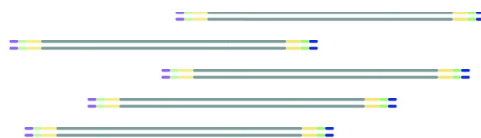
vs



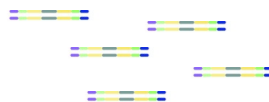
Length



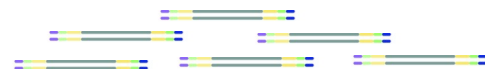
vs



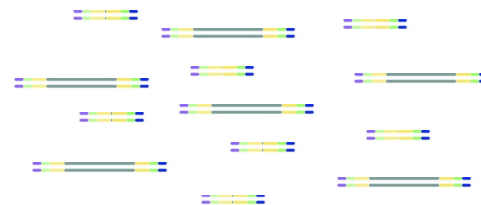
or



Remove adapter dimers



vs

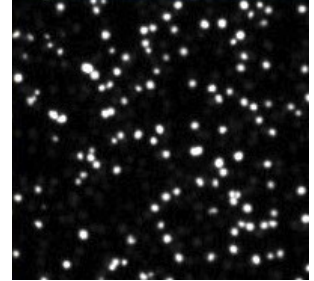


Library prep QC - Concentration

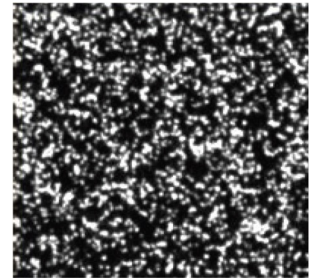
Why is an accurate concentration important?

- Individual samples: to pool equal amounts of different samples together
 - Aim for an even amount of sequencing data per sample
- Final library: sequencer requires loading with a very precise concentration
 - If concentration is overestimated (actual conc. is lower than what you think)
 - Underloading and underclustering of sequencing run
 - Sequencing run will yield less sequencing data than usual
 - If concentration is underestimated (actual conc. is higher than what you think)
 - Overloading and overclustering of sequencing run
 - Sequencing run will suffer from higher error rates and reduced data output

Underloaded



Overloaded



Library prep QC - Concentration

DO NOT USE to quantify libraries

Spectrophotometry
(e.g. Nanodrop)



- DNA absorbs light most strongly at 260 nm
- Level of absorbance ~ concentration of DNA

BUT other molecules (RNA, proteins) & contaminants also absorb light at 260 nm, causing up to 10-fold **overestimation** of the DNA concentration!

Fluorometry
(e.g. Qubit, qPCR)



- Detects fluorescent dyes that only bind DNA
- Most reliable method for NGS

Electrophoresis
(e.g. Tapestation, Bioanalyzer)



- Compare intensity of DNA bands to standards
- Especially useful to determine concentration in certain fragment length range

Library prep QC - Concentration

Qubit dsDNA High Sensitivity (HS) Assay kit



1. Prepare Working Solution:

- Diluting HS reagent 1:200 in HS buffer

1. Dilute in Working Solution:

- Standards: 10 μ l standard + 190 μ l Working Solution

- dsDNA HS Reagent (200x concentrate)
- Samples: 1 μ l sample + 199 μ l Working Solution

- dsDNA HS Buffer

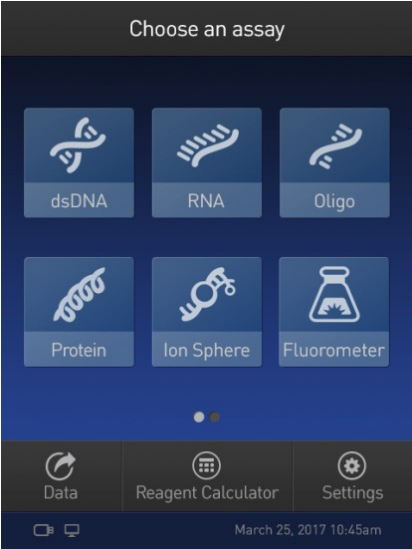
- dsDNA HS Standard #1

- dsDNA HS Standard #2

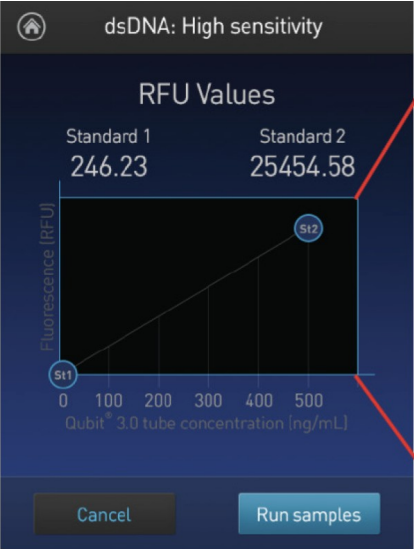
1. Vortex 2-3 seconds before running standards/samples

Library prep QC - Concentration

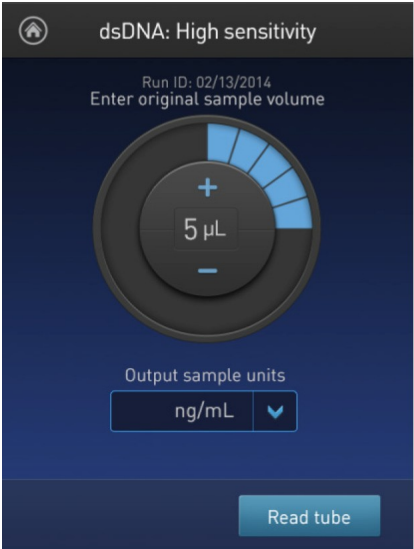
Qubit dsDNA High Sensitivity (HS) Assay kit



Select dsDNA HS workflow



Run the 2 standards



Enter sample volume
added to tube



Run sample
Top value = original conc.
Bottom value = diluted conc.

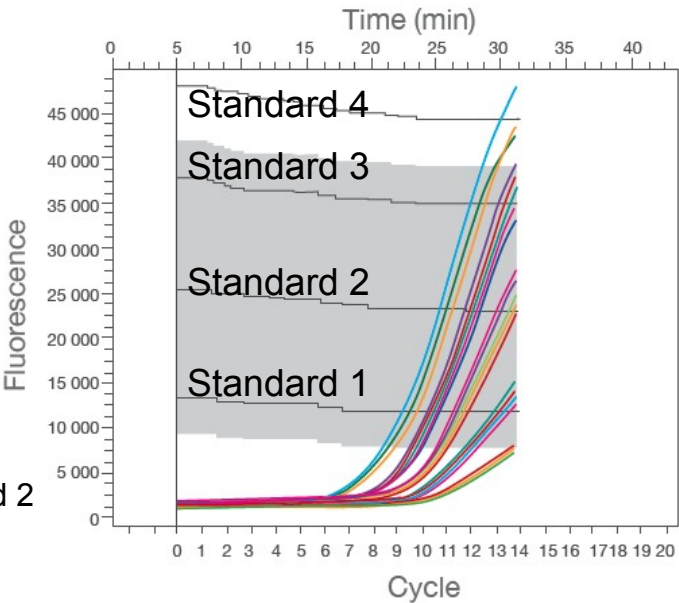
Library prep QC - Concentration

- **What if the concentration is too low?**
 - In future runs, consider adding more cycles to the indexing PCR step
 - Amplify library using P5/P7 primers and qPCR-based library amplification kit
 - e.g. Roche KK2702

1. Set up qPCR reaction: library + mastermix + P5/P7 primers
2. Treat standards as separate samples (do not add mastermix)
3. Start run

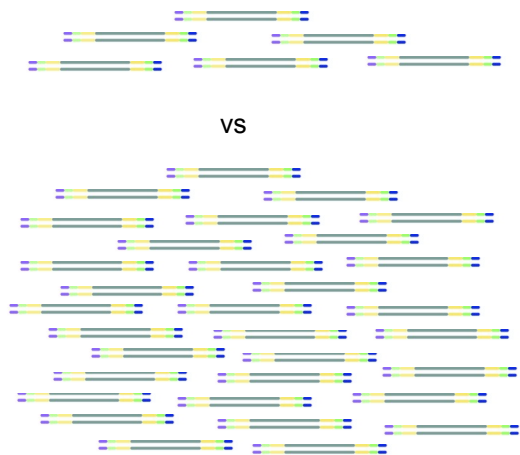
Temperature	Time	Cycles
98°C	45 sec	1
98°C	15 sec	25
63°C	30 sec	
72°C	1m 45 sec	
Plate read		
72°C	20 sec	

1. Pause cycler during the 20s at 72°C step when a sample crossed the Standard 2 line, remove the sample and continue amplification for other samples
2. After amplification, perform bead cleanup on amplified libraries and QC again

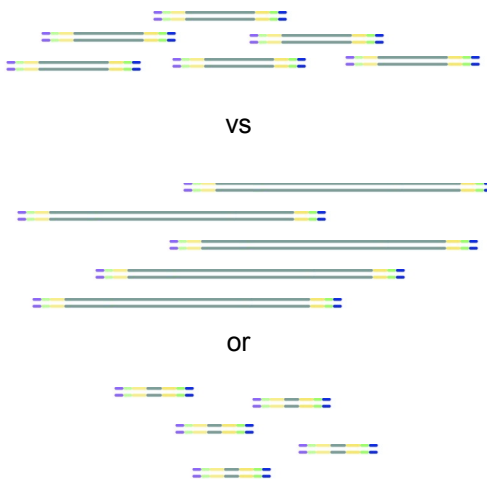


Library prep QC

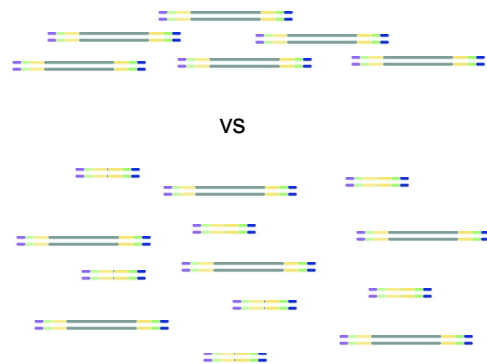
Concentration



Length



Remove adapter dimers



Library prep QC - Length

Why is it important to assess fragment length in a library?

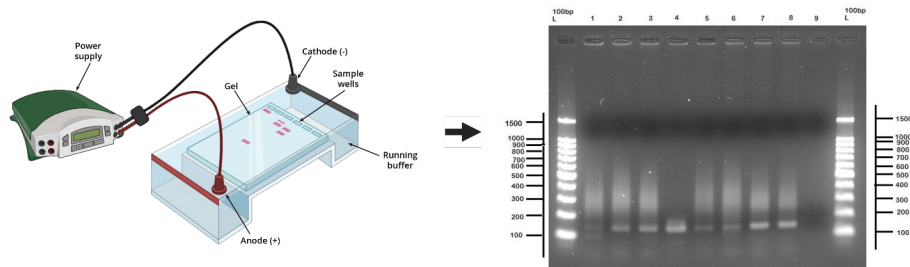
- Long fragments (>1.2 kb) do not cluster on Illumina flow cells and are therefore not sequenced
- Short fragments are preferentially amplified (better clustering) and will thus be preferentially sequenced
- Fragment length distribution impacts calculations to go from ng/μl (Qubit) to nM for loading concentrations

$$\text{Concentration in nM} = \frac{\text{concentration in ng/}\mu\text{l}}{660\text{g/mol} \times \text{average library size in bp}} \times 10^6$$

Qubit
↓
Capillary or gel electrophoresis
↑

Library prep QC - Length

Gel electrophoresis

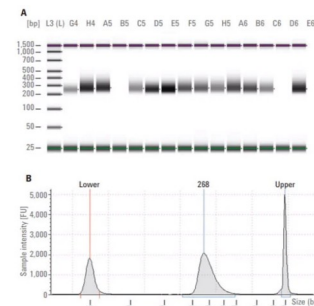


1. PCR amplify an aliquot of the library using universal P5/P7 Illumina primers
2. Gel electrophoresis of ladder and samples

- Takes ~ 4 hrs

Capillary electrophoresis

E.g. Tapestation, Bioanalyzer



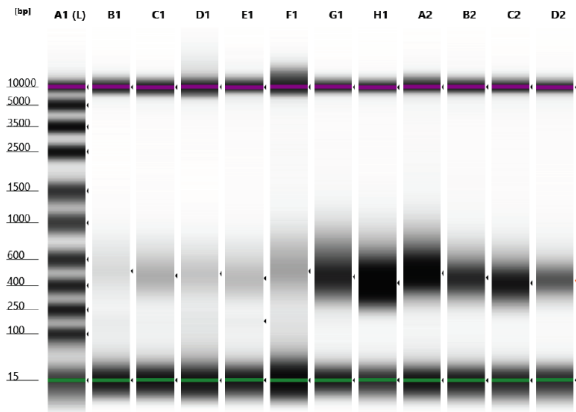
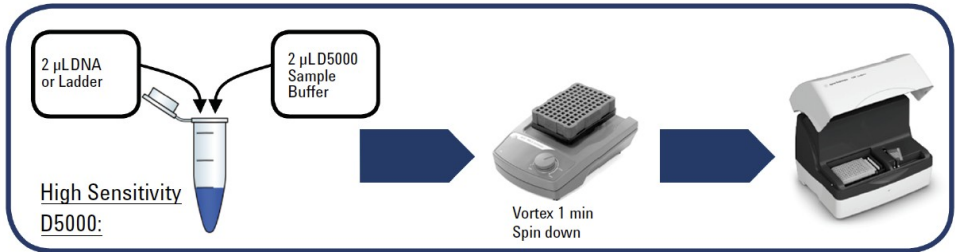
1. Add buffer to aliquot of library
2. Run machine

- Tapestation ~ 10 minutes (depending on # samples)
- Bioanalyzer ~ 45 minutes
- No PCR bias
- Can calculate concentration in nM

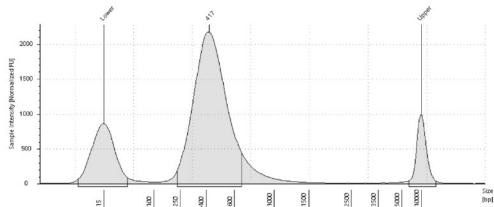
Library prep QC - Length

Tapestation

- High Sensitivity D1000 or D5000 kits (1kb or 5kb detection limit)



Peak: H1

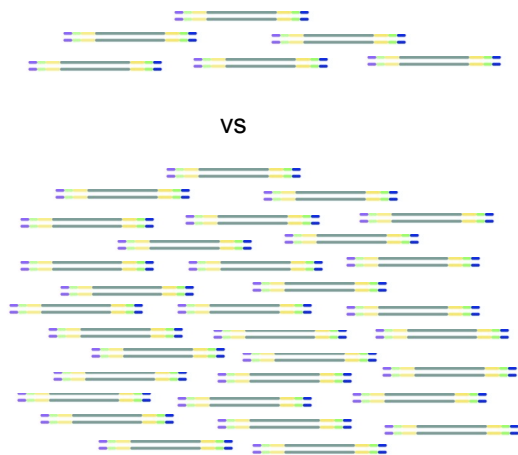


Location	H1
Concentration	1470
Description	
Observations	

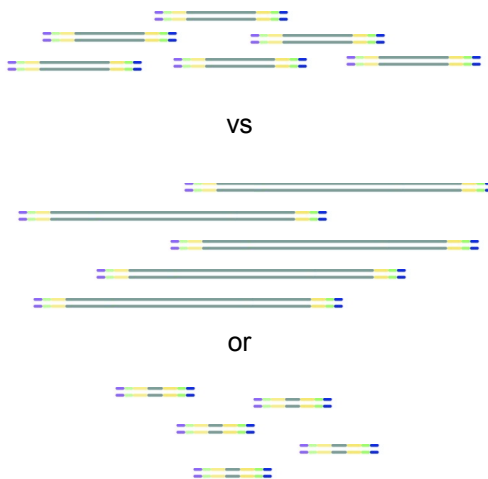
Size [bp]	Calibrated Conc. [pg/ul]	Assigned Conc. [pg/ul]	Peak Molarity [µmM]	% Integrated Area	Peak Comment	Observations
15	346	-	35500	-		Lower Marker
417	1470	-	5430	100.00		
10000	180	180	27.7	-		Upper Marker

Library prep QC

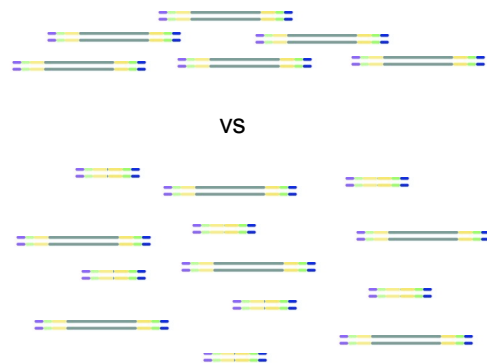
Concentration



Length



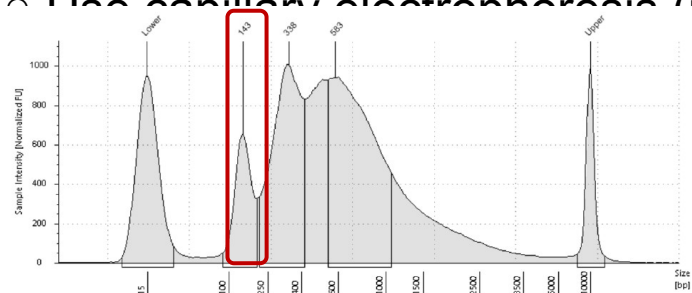
Remove adapter dimers



Library prep QC - Clean up adapter dimers

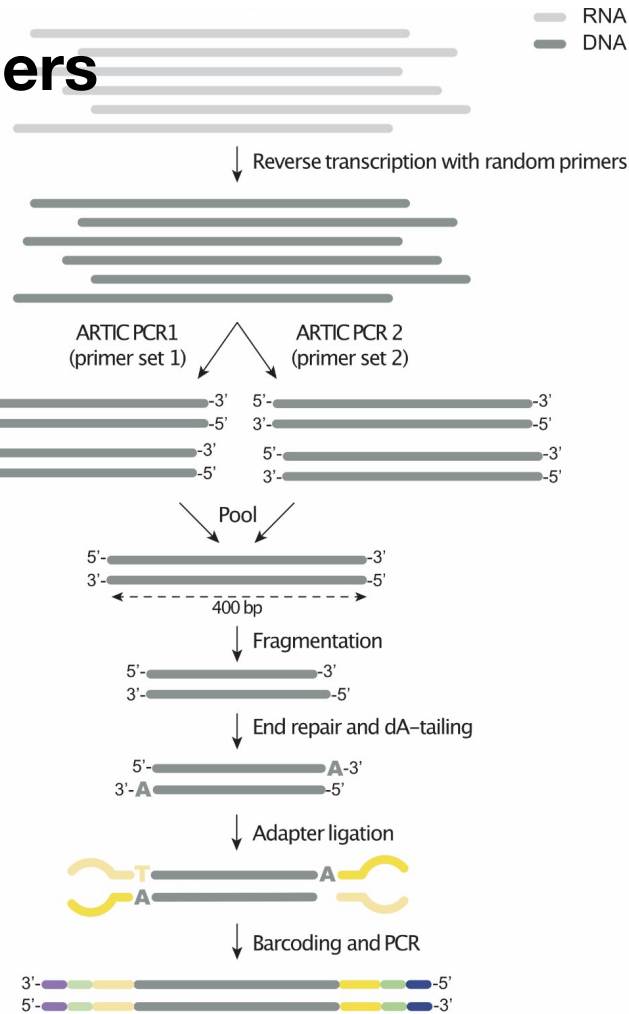
- Adapter dimers are:
 - A byproduct of the adapter ligation step during library prep (TruSeq/NEBNext Ultra II DNA kits)
 - ± 140 bp long at the final QC step (after barcoding PCR)

○ Use capillary electrophoresis (preferred) or



Adapter dimers

detect adapter
Adapter dimer

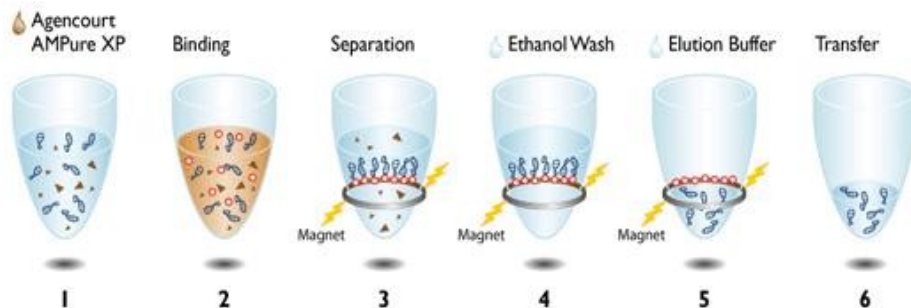
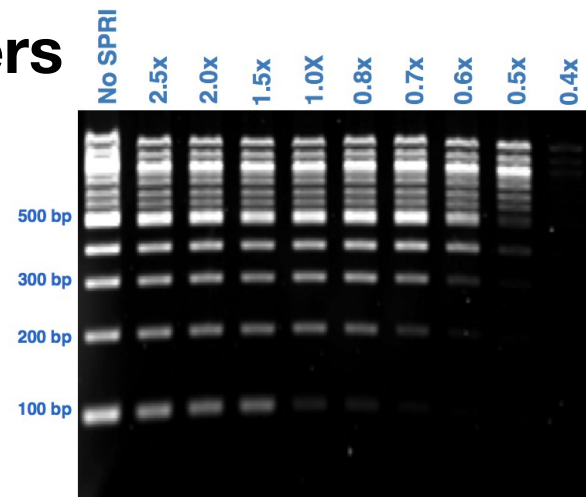


Library prep QC - Clean up adapter dimers

AMPure XP SPRI select beads (paramagnetic)

or homemade SPRI beads (>10x cheaper)

- The longer a fragment, the better it will bind to the beads
- The fewer beads you add, the more short fragments you'll lose
- **Perform size selection using these beads to remove adapter dimers**



1. PCR reaction 2. Binding of PCR amplicons to magnetic beads 3. Separation of PCR amplicons bound to magnetic beads from contaminants 4. Washing of PCR amplicons with Ethanol 5. Elution of PCR amplicons from the magnetic particles 6. Transfer away from the beads into a new plate

$$\text{SPRI ratio} = \frac{\text{Volume beads}}{\text{Volume reaction}}$$

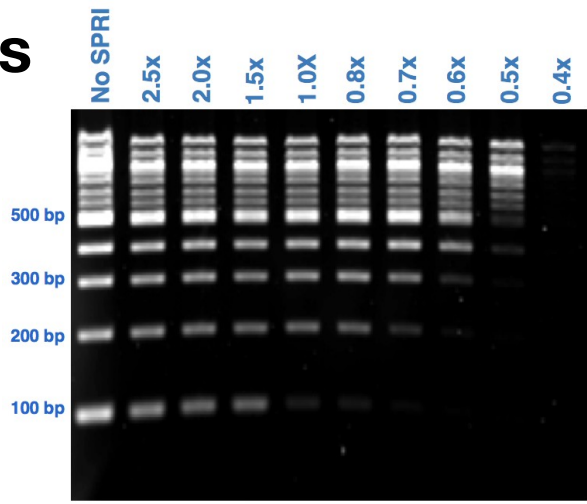
Example: for a 1.8x SPRI clean-up:

- Add 18 μl of SPRI beads to 10 μl of sample

Library prep QC - Clean up adapter dimers

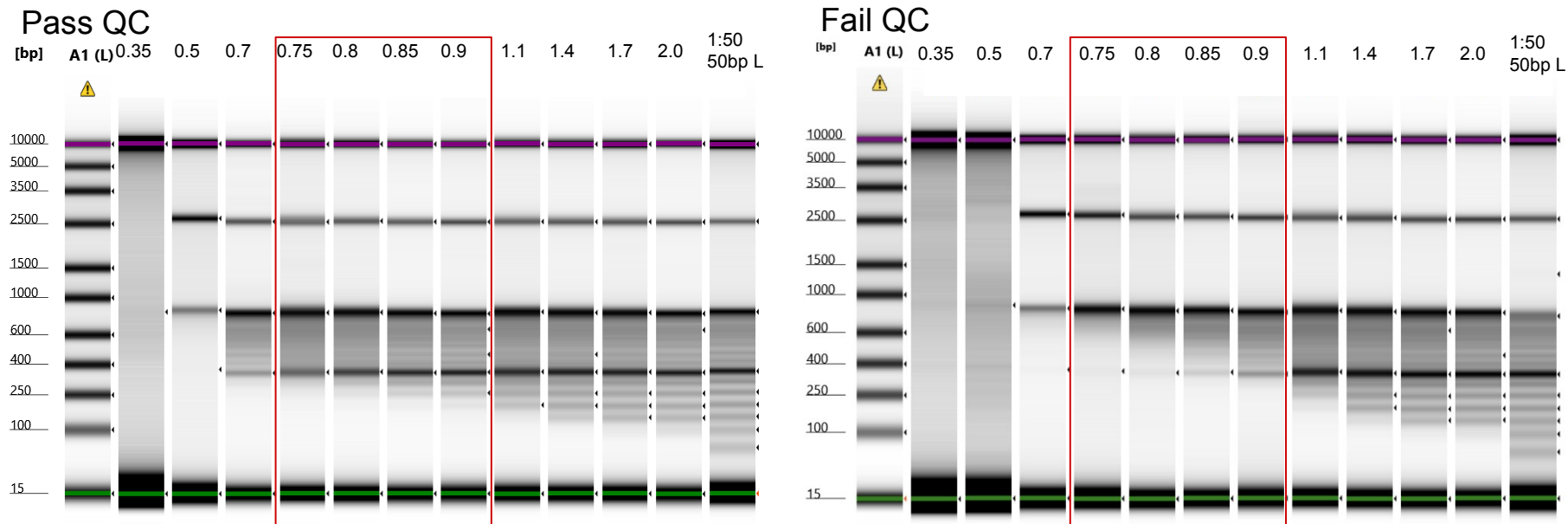
AMPure XP SPRI select beads (paramagnetic)

- Size selection using beads to remove adapter dimers
- Rule of thumb: if you see a clear trace of adapter dimers, performs another round of cleanup



Step	To remove	SPRI ratio
After second strand synth.	Reagents	1.8x
After adapter ligation	Adapter dimers (64 bp) & reagents	0.9x
After P7/P5 PCR	P5-adapter dimer-P7 (143 bp) & reagents	0.8x
QC of library	P5-Adapter dimer-P7 (143 bp)	0.8x

You can QC any batch of beads using a 50bp ladder

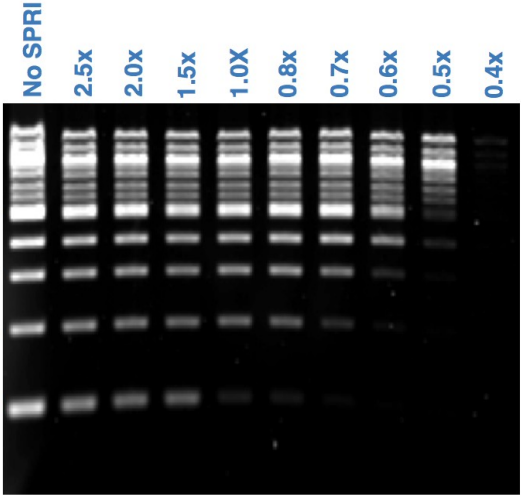


- Expired beads or beads where proper storage is uncertain are good candidates for a quick QC check
- [Appendix E, part II](#) for QC protocol. (Do not follow part I - we no longer recommend homemade beads)

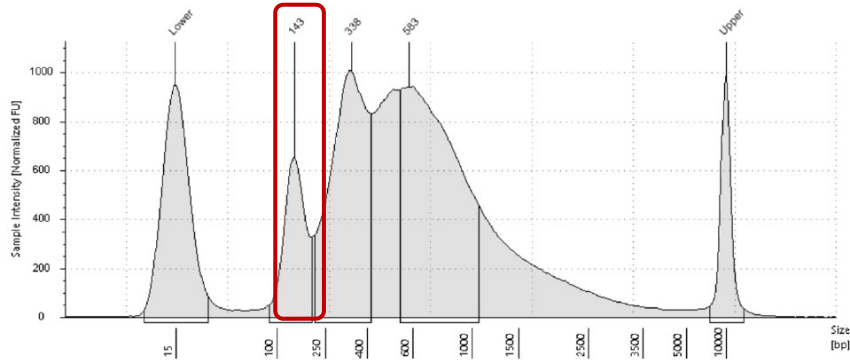
Library prep QC - Clean up adapter dimers

AMPure XP SPRI select beads (paramagnetic)

- Size selection using SPRI beads to remove adapter dimers
- Rule of thumb: if you see a clear trace of adapter dimers, performs another round of cleanup



After PCR amplification

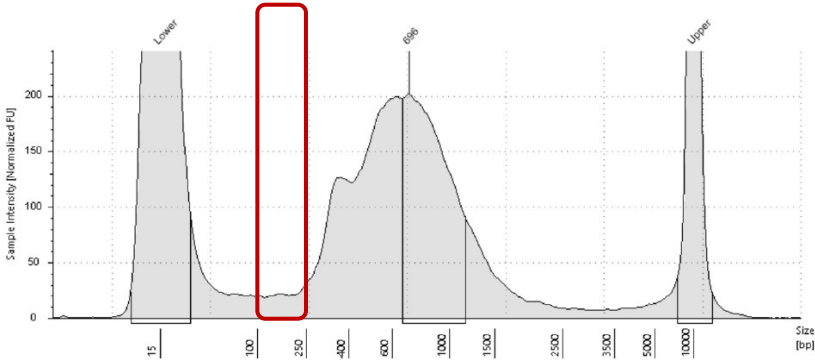


Adapter dimers

3 rounds
SPRI
(0.85x)

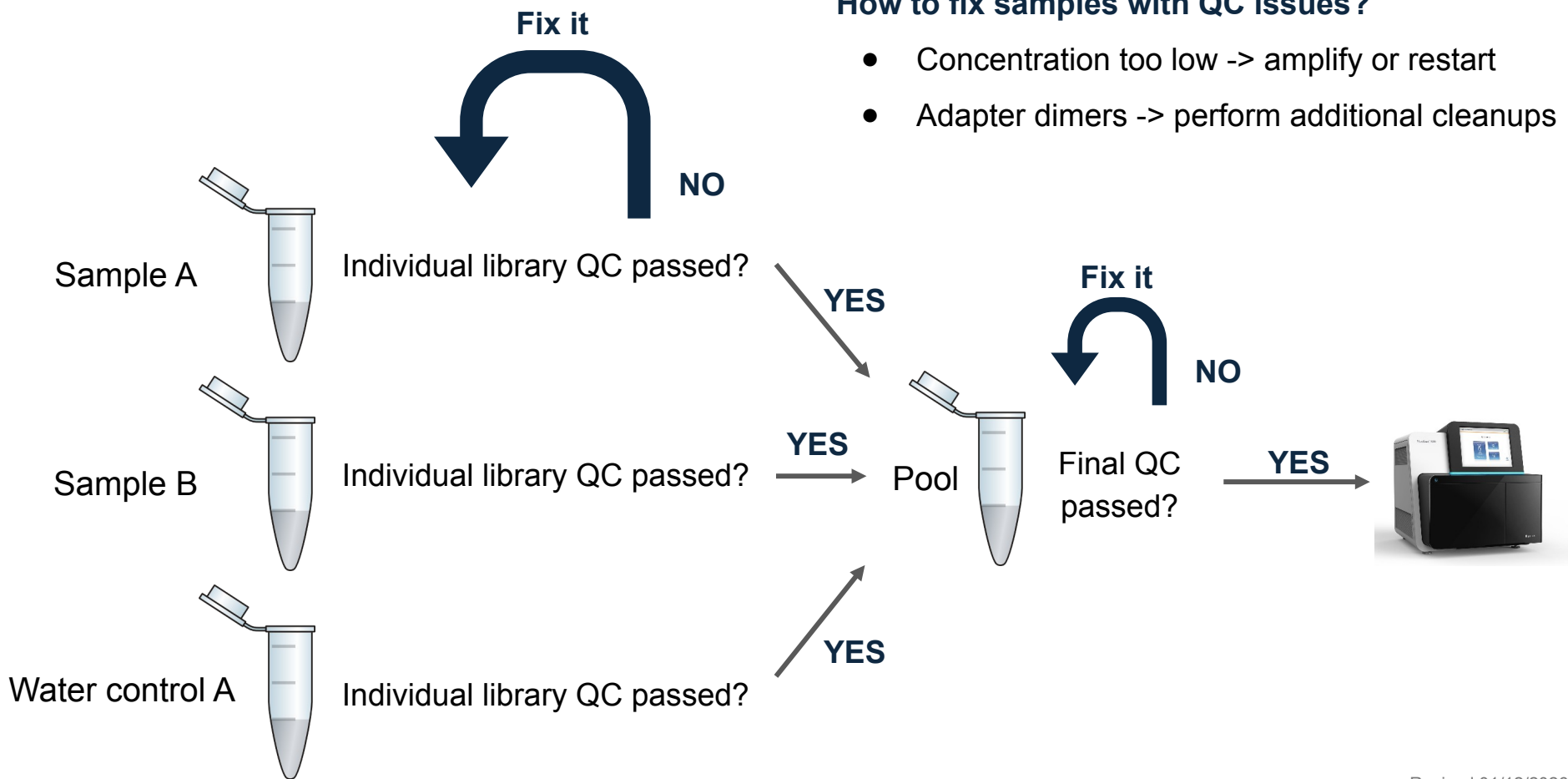


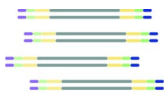
After SPRI bead clean-up



ALL DIMERS SHOULD BE REMOVED!

Library prep QC

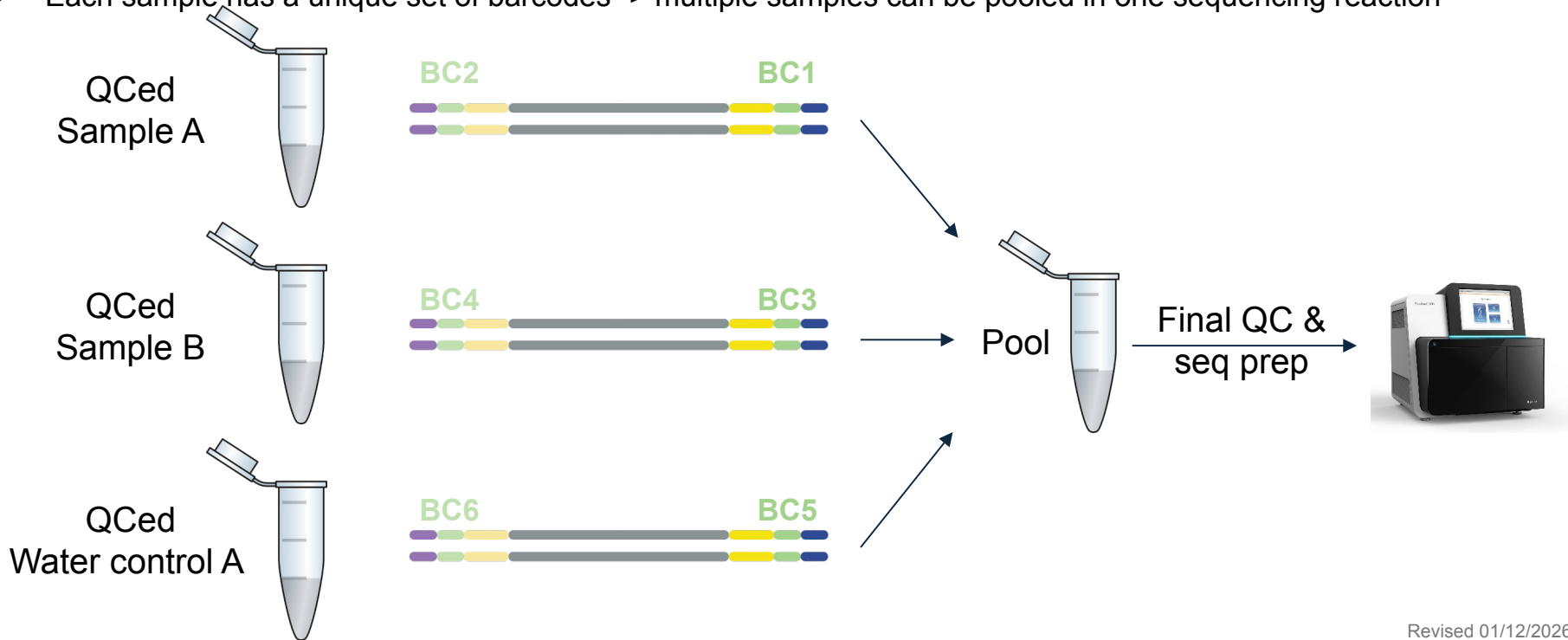


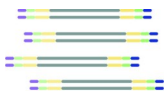


Library prep QC

Pooling samples for sequencing

- Each sample has a unique set of barcodes -> multiple samples can be pooled in one sequencing reaction





Library prep QC

Pooling samples together for sequencing

1. Qubit to determine concentration of individual libraries (expect library to be 0.5 to 10 ng/ul)
2. Pool barcoded libraries so that all are equally represented (except H₂O controls, only 1:5)

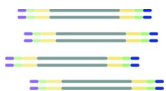


<u>Qubit concentration</u>			<u>Pool</u>		
Sample 1:	10 ng/ul	->	4.0 µl	->	40 ng
Sample 2:	5 ng/ul	->	8.0 µl	->	40 ng
Sample 3:	8 ng/ul	->	5.0 µl	->	40 ng
Sample 4:	15 ng/ul	->	2.6 µl	->	40 ng
H ₂ O control:	1 ng/ul	->	8.0 µl	->	8 ng
			Total:	27.6 µl	168 ng

End goal is >10 µl of 2 nM pooled library (or 4 nM for MiSeq)

Pooling guidelines

- Do not pool more than half of your library volume (in case you need to go back)
- Pool waters/negative controls less mass than samples (1/5th or 1/10th)



Library prep QC

Pooling samples together for sequencing

3. After pooling, determine quantity and quality of pool

Pick one way to calculate nM and stick with it!

- Qubit concentration and library size
or
- Automatic calculation by Tapestation

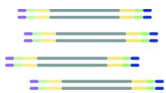
$$\text{Concentration in nM} = \frac{\text{concentration in ng/}\mu\text{l}}{660\text{g/mol} \times \text{average library size in bp}} \times 10^6$$

Qubit
↓

↑
Capillary or gel
electrophoresis

Example: if concentration is 6.1 ng/μl, and average library size is 550 bp -> concentration is 16.8 nM

End goal is >10 μl of 2 nM pooled library



Library prep - preparing to load the sequencer

Dilute library to the correct loading concentration:

- a. iSeq:
 - i. dilute to 75 pM (or another optimized concentration)
 - ii. Spike in 1-5% of PhiX at the same concentration

- a. MiSeq:
 - i. First denature library with sodium hydroxide (NaOH)
 - ii. Then dilute to 10-12 pM (or another optimized concentration)
 - iii. Spike in 1-5% of denatured PhiX to denatured and diluted library

See slide decks for iSeq or MiSeq loading for more details!